Discrete Random Variables

Section 1: Probability Distribution Tables & Total Probability

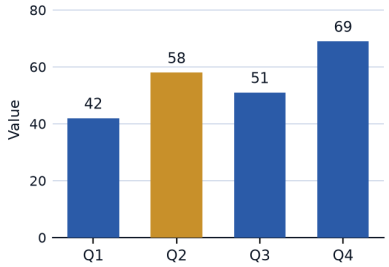
1 The random variable X has probability distribution $P(X = x) = c(x^2 + 1)$ for $x = 0, 1, 2, 3$. Find the constant c.

a. $\frac{1}{18}$

b. $\frac{1}{14}$

c. $\frac{1}{15}$

d. $\frac{1}{20}$



2 A fair die is rolled. Let X be the number of rolls until a 6 appears. What kind of distribution does X follow?

a. Geometric distribution

b. Binomial distribution

c. Uniform distribution

d. Poisson distribution

3 Given the probability distribution of Y: $P(Y = -1) = 0.3, P(Y = 0) = 0.4, P(Y = 2) = 0.3$. Find $P(Y \geq 0)$.

a. 0.7

b. 0.4

c. 0.3

d. 1.0

Section 2: Expected Value $E(X)$ & Moments

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4 Find the expectation $E(X)$ for the distribution:
$P(X=1)=0.1, P(X=2)=0.3, P(X=3)=0.4, P(X=4)=0.2$.

a. 2.7

b. 2.5

c. 3.0

d. 2.8

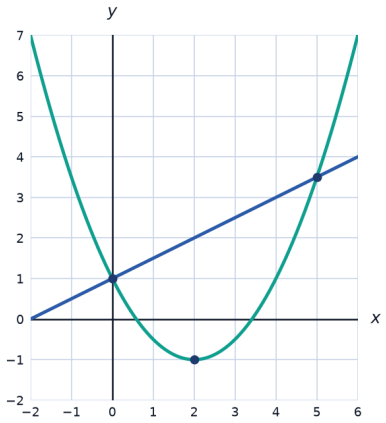
5 If $E(X)=5$, what is $E(3X-4)$?

a. 11

b. 15

c. 19

d. 7



6 A game charges 5 dollars to play. A fair coin is tossed: Heads pays 8 dollars, Tails pays nothing. What is the expected net gain for the player?

a. -$1.00

b. +$1.00

c. -$2.00

d. +$4.00

Section 3: Variance $\mathrm{Var}(X)$ & Standard Deviation

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7 Given $E(X) = 3$ and $E(X^2) = 13$, find $\text{Var}(X)$.

a. 4

b. 10

c. 2

d. 16

8 If $\text{Var}(X) = 6$, find the variance of $Y = 2X - 5$.

a. 24

b. 12

c. 7

d. 36

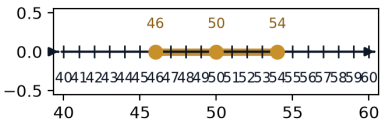
9 Calculate the standard deviation of X given $P(X = 0) = 0.5$ and $P(X = 2) = 0.5$.

a. 1

b. 2

c. 0.5

d. $\sqrt{2}$



Section 4: Multi-Stage Discrete Problem Solving

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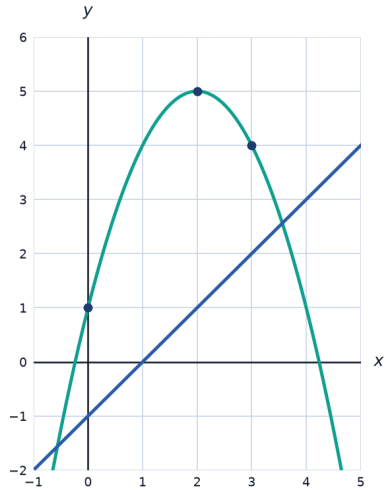
10 Two fair 4-sided dice (numbered 1 to 4) are rolled. Let S be the sum of the two numbers. Find $P(S > 5)$.

a. $\frac{6}{16} = \frac{3}{8}$

b. $\frac{1}{2}$

c. $\frac{5}{16}$

d. $\frac{1}{4}$



11 For the sum S of the two 4-sided dice in the previous question, what is $E(S)$?

a. 5

b. 4

c. 6

d. 4.5

12 A box has 3 red and 2 blue balls. Two balls are drawn without replacement. Let X be the number of blue balls drawn. Find $\mathrm{Var}(X)$.

a. 0.36

b. 0.40

c. 0.48

d. 0.24

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